

Solution Requirements

Project Title

Electricity Consumption Analysis Dashboard Using Tableau

1. Introduction

The Electricity Consumption Analysis Dashboard is designed to analyze electricity usage across different states, regions, sectors, and years. The project provides interactive visualizations that help users understand consumption patterns and support better decision-making through Tableau dashboards and stories.

2. Functional Requirements

Functional Requirement

- Import electricity consumption dataset into Tableau.
 - Display state-wise electricity consumption using Bar Chart.
 - Display region-wise electricity consumption.
 - Display sector-wise electricity consumption using Pie Chart.
 - Display monthly electricity consumption trend using Line Chart.
 - Display Heat Map for State vs Sector analysis.
 - Provide interactive filters (Year, State, Region, Sector).
 - Create Dashboard containing all visualizations.
 - Create Story explaining the analysis step-by-step.
 - Publish Dashboard and Story on Tableau Public.
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3. Non-Functional Requirements

- Dashboard should load quickly.
- Visualizations should be interactive and easy to understand.
- Dashboard should be available online through Tableau Public.
- Data should be accurate and reliable.

- Dashboard should be user-friendly.
- Filters should respond efficiently.
- Dashboard should support future data updates.
- Reports should be visually attractive and informative.

4. Software Requirements

Software	Purpose
Tableau Desktop	Dashboard Development
Tableau Public	Dashboard Publishing
Microsoft Excel	Dataset Preparation
Visual Studio Code	Flask Website Development
Python 3.x	Flask Backend
Flask	Web Application Framework
GitHub	Project Repository

5. Hardware Requirements

Hardware	Specification
Processor	Intel Core i3 / AMD Ryzen 3 or above
RAM	Minimum 4 GB (8 GB Recommended)
Storage	500 MB Free Space
Operating System	Windows 10/11

6. Expected Outcome

The proposed solution will provide an interactive dashboard that enables users to analyze electricity consumption efficiently through charts, maps, filters, and stories. It simplifies data interpretation and supports informed decision-making for electricity management.
